

Activity: Visualize CNN Filters

Introduction

In this activity, you'll explore how Convolutional Neural Networks (CNNs) "see" by extracting and analyzing filters from a pre-trained **VGG16 model**. You'll visualize the learned filters from various layers, generate feature maps to understand the patterns the network detects, and create Grad-CAM heatmaps to reveal which regions of an image influence the model's decisions.

Learning Outcomes

By the end of this activity, you will:

1. Load and preprocess an image for use with a pre-trained **VGG16 model**.
2. Extract and visualize **filters** from different convolutional layers to understand their learned patterns.
3. Generate and interpret **feature maps** to see how CNNs detect patterns like edges or textures.
4. Create and analyze **Grad-CAM heatmaps** to identify regions critical to the model's classification decisions.

Working with Dev Container

To complete this assignment in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Starter File

- Open the notebook: `activity-visualize-cnn-filters.ipynb` in the `Code` folder.
- Run it locally or in **Google Colab**.
- Ensure the following libraries are installed: `tensorflow`, `opencv-python`, `numpy`, `matplotlib`.
- Use the provided image in `images` under name of `horse.jpg`.
- The pre-trained **VGG16 model** is loaded directly from TensorFlow's `keras.applications` module.

Code Steps

1. Load and Preprocess the Image
2. Extract and Visualize Filters from Convolutional Layers
3. Generate and Visualize Feature Maps
4. Create and Analyze Grad-CAM Heatmaps

Conclusion

In this activity, you successfully:

- Loaded and preprocessed an image for the **VGG16 model**.
- Extracted and visualized **filters** from early and deep convolutional layers, observing the progression from edges to complex patterns.
- Generated and interpreted **feature maps**, identifying which image regions activate specific filters.
- Created **Grad-CAM heatmaps** to reveal the image regions critical to the model's classification decisions.